LIST OF EXPERIMENTS

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1.		Develop the test plan for testing an e-commerce web / mobile application (www.amazon.in).			
2.		Design the test cases for testing the e- commerce application.			
3.		Test the e-commerce application and report the defects in it.			
4.		Develop the test plan and design the test cases for an inventory control system.			
5.		Execute the test cases against a client server or desktop application and identify the defects.			
6.		Test the performance of the e-commerce application.			
7.		Automate thetesting of e-commerce applications using Selenium.			
8.		Integrate TestNG with the above test automation.			

EX.NO: 1	**Develop the test plan for testing an e-commerce web/mobile application (www.amazon.in)**

AIM:

The aim of this experiment is to develop a comprehensive test plan for testing the functionality and usability of the e-commerce web/mobile application www.amazon.in.

ALGORITHM:

Test Plan: www.amazon.in E-commerce Application

1. **Introduction**

•**Purpose:** To ensure the quality, reliability, and performance of the www.amazon.in e-commerce web and mobile application.

•**Scope:** This test plan covers both web and mobile versions of the application.

•***Testing Objectives:***

•Validate the functionality of the application.

•Ensure usability and user experience.

•Evaluate security and performance.

•Verify compatibility with various browsers and mobile devices.

•**References:** Requirements document, design specifications, user stories.

2. **Test Strategy**

•**Testing Approach:** A comprehensive testing approach combining functional and non-functional testing.

•***Test Environment:***

•**Web:** Latest versions of Chrome, Firefox, Safari, and Edge on Windows and macOS.

•**Mobile:** iOS (latest) and Android (latest) devices.

•***Test Data:***

•For user accounts: A set of predefined test accounts with various roles (admin, customer).

•For products: A variety of sample products for testing.

3. **Test Phases**

•**Unit Testing:** Test individual components and functions within the application.

•**Integration Testing:** Test the integration of components and modules.

•**System Testing:** Test the complete system to ensure it meets all requirements.

•**User Acceptance Testing (UAT):** Involve end-users to validate usability and functionality.

4. **Test Scenarios**

- ***List of Test Scenarios****:*
  - User registration and authentication.
  - Product searching and browsing.
  - Adding/removing items from the shopping cart.
  - Payment processing.
  - Order tracking and history.
  - User reviews and ratings.
  - Mobile-specific features (e.g., push notifications).

5. **Test Cases**

- ***Detailed test cases for each scenario****:*
  - Test ID.
  - Description.
  - Precondition.
  - Steps to Reproduce.
  - Expected Results.
  - Actual Results.
  - Pass/Fail status.

6. **Test Execution**

- *Test execution approach:*
  - Manual testing for functional and usability testing.
  - Automation for regression testing.
- Testing schedule.
- Test environments.
- Roles and responsibilities of the testing team members.

7. **Defect Management**

- **Defect tracking tool:** Use JIRA.
- Defect reporting and documentation process.
- Defect prioritization and severity criteria.

8. **Test Deliverables**

List of documents and artifacts:

- Test plan.
- Test cases.
- Test reports

Defect logs.

9. **Risk Assessment**

- ***Identify potential risks****:*
  - Data security.
  - Scalability issues.
  - Usability challenges.
- Mitigation strategies for identified risks.

10. **Schedule**

Preparation (1 week)

- Develop test plan and cases.
- Set up environments.

Testing (5 weeks)

- Unit, Integration, System.
- UAT (2 weeks).

Automation (2 weeks)

Closure (1 week)

- Documentation and approvals.

11. Exit Criteria

- *Conditions to conclude testing:*
  - 95% of test cases passed.
  - All critical defects resolved.

12. Approvals

- Stakeholders for test plan review and approval: Project Manager, Development Lead, QA Lead.

RESULT:

Upon completion of the experiment, you will have a well-structured test plan that provides a clear road map for testing the e-commerce web/mobile application www.amazon.in.

EX.NO: 2	**Design the test cases for testing the e-commerce application.**

AIM:

The aim of this experiment is to design a set of comprehensive and effective test cases for testing the e- commerce application www.amazon.in.

ALGORITHM:

1. **Understand Requirements:** Familiarize yourself with the functional and non-functional requirements of the e-commerce application.
2. **Identify Test Scenarios:** Based on the requirements, identify different test scenarios that cover all aspects of the application.
3. **Write Test Cases:** Develop test cases for each identified scenario, including preconditions, steps to be executed and expected outcomes.
4. **Cover Edge Cases:** Ensure that the test cases cover edge cases and boundary conditions to verify the robustness of the application.
5. **Prioritize Test Cases:** Prioritize the test cases based on their criticality and relevance to the application.
6. **Review Test Cases:** Conduct a peer review of the test cases to ensure their accuracy and completeness.
7. **Optimize Test Cases:** Optimize the test cases for reusability and maintainability.

TEST CASE DESIGN:

The test case design should include the following components for each test case:

- **Test Case ID:** A unique identifier for each test case.
- **Test Scenario:** Description of the scenario being tested
- **Test Case Description:** Detailed steps to execute the test.
- **Precondition:** The necessary conditions that must be satisfied before executing the test case.
- **Test Steps:** The sequence of actions to be performed during the test.
- **Expected Result:** The outcome that is expected from the test.

Test Case Table:

Process	No.	Test Case	Steps	Description	Status	Expected Result	Actual Result	Comment
Test Case Design	TC001	User Registration	1. Navigate to the registration page.	Verify user registration process.	Done	User can successfully register.		
	TC002	User Login	1. Navigate to the login page.	Verify user login process.	Done	User can successfully log in.		
	TC003	Search Functionality	1. Enter a keyword in the search bar.	Verify search functionality.	Done	Search results relevant to the keyword.		
	TC004	Add to Cart	1. Browse the product catalog.	Verify adding products to the cart.	Done	Product is added to the shopping cart.		
	TC005	Shopping Cart Validation	1. Click on the shopping cart icon.	Verify the shopping cart contents.	Done	Items in the shopping cart are displayed.		

EXPLANATION:

Test cases are designed to validate the functionality and behavior of the e-commerce application. They ensure that the application performs as intended and meets the specified requirements.

RESULT:

Upon completion of the experiment, you will have a set of well-defined test cases ready for testing the e-commerce application www.amazon.in.

EX.NO: 3	Test the e-commerce application and report the defects in it.

AIM:

The aim of this experiment is to execute the designed test cases and identify defects or issues in the e-commerce application www.amazon.in.

ALGORITHM:

1. **Test Environment Setup:** Set up the testing environment with the required hardware, software, and test data.
2. **Test Case Execution:** Execute the test cases designed in Experiment 2, following the specified steps.
3. **Defect Identification:** During test execution, record any discrepancies or issues encountered.
4. **Defect Reporting:** Log the identified defects with detailed information, including steps to reproduce, severity, and priority.
5. **Defect Tracking:** Track the progress of defect resolution and verify fixes as they are implemented.
6. **Retesting:** After defect fixes, retest the affected areas to ensure the issues are resolved.
7. **Regression Testing:** Conduct regression testing to ensure new changes does not introduce new defects.

Test Case Table:

Process	No.	Test Case	Steps	Description	Status	Expected Result	Actual Result	Comment
Test Case Design	TC001	User Registration	1. Navigate to the registration page.	Verify user registration process.	Done	User can successfully register.		
	TC002	User Login	1. Navigate to the login page.	Verify user login process.	Done	User can successfully log in.		
	TC003	Search Functionality	1. Enter a keyword in the search bar.	Verify search functionality.	Done	Search results relevant to the keyword.		
	TC004	Add to Cart	1. Browse the product catalog.	Verify adding products to the cart.	Done	Product is added to the shopping cart.		

EXPLANATION:

Testing the e-commerce application aims to validate its functionality and usability. By identifying and reporting defects, you ensure the application's quality and reliability.

RESULT:

Upon completion of the experiment, you will have a list of identified defects and their status after resolution.

EX.NO: 4	**Develop the test plan and design the test cases for an inventory control system.**

AIM:

The aim of this experiment is to create a comprehensive test plan and design test cases for an Inventory Control System.

ALGORITHM:

1. **Identify the Scope:** Determine the scope of testing, including the features and functionalities that need to be tested.
2. **Define Test Objectives:** Specify the primary objectives of testing, such as functional testing, usability testing, performance testing, security testing, etc.
3. **Identify Test Environment:** Define the platforms, browsers, devices, and operating systems on which the application will be tested.
4. **Determine Test Deliverables:** Decide on the documents and artifacts that will be generated during the testing process, such as test cases, test reports, and defect logs.
5. **Create Test Strategy:** Develop an overall approach for testing, including the testing techniques, entry and exit criteria, and the roles and responsibilities of the testing team.
6. **Define Test Scope and Schedule:** Specify the timeline for each testing phase and the scope of testing for each phase.
7. **Risk Analysis:** Identify potential risks and their impact on the testing process, and device risk mitigation strategies.
8. **Resource Planning:** Allocate the necessary resources, including the testing team, hardware, and software required for testing.
9. **Test Case Design:** Prepare detailed test cases based on the requirements and functionalities of the e- commerce application.
10. **Test Data Setup:** Arrange test data required for executing the test cases effectively.
11. **Test Execution:** Execute the test cases and record the test results.
12. **Defect Reporting:** Document any defects encountered during testing and track their resolution.

TEST PLAN:

The test plan should cover the following sections:

1. **Introduction:** Briefly describe the purpose of the test plan and provide an overview of the e- commerce application to be tested.
2. **Test Objectives:** List the primary objectives of testing the application.
3. **Test Scope:** Specify the features and functionalities to be tested and any limitations on testing.
4. **Test Environment:** Describe the hardware, software, browsers, and devices to be used for testing.
5. **Test Strategy:** Explain the overall approach to be followed during testing.
6. **Test Schedule:** Provide a detailed timeline for each testing phase.

Test Plan:

Process	No.	Test Case	Steps	Description	Status	Expected Result	Actual Result	Comment
Test Plan	TC001	Scope of Testing	1. Review the requirements and project documentation.	Verify the scope of testing.	Done	The test plan includes all essential features.		
			2. Identify the modules to be tested.					
			3. Determine the out-of-scope items.					

Process	No.	Test Case	Steps	Description	Status	Expected Result	Actual Result	Comment
	TC002	Test Objectives	1. Review the requirements and project documentation.	Verify the test objectives.	Done	The test objectives are clearly defined.		
			2. Discuss with stakeholders to understand expectations.					
	TC003	Test Environment	1. Identify the hardware and software requirements.	Verify the required environments.	Not Started	The test environment is defined.		
			2. Set up the required hardware and software.					
	TC004	Test Deliverables	1. Determine the documents and artifacts to be produced.	Verify the required deliverables.	Not Started	All necessary documents are listed.		
			2. Create templates for test reports, defect logs, etc.					

Test Case Design:

Process	No.	Test Case	Steps	Description	Status	Expected Result	Actual Result	Comment
Test Case Design	TC001	Module A - Functionality Test	1. Review the requirements related to Module A.	Verify the functionality of Module A.	Not Started	All functionalities of Module A are tested.		
			2. Identify test scenarios for Module A.					
			3. Develop detailed test cases for Module A.					
	TC002	Module B - Integration Test	1. Review the requirements related to Module B.	Verify the integration of Module B with others.	Not Started	Module B is successfully integrated.		
			2. Identify integration points with other modules.					

Process	No.	Test Case	Steps	Description	Status	Expected Result	Actual Result	Comment
	TC003	Module C - Performance Test	1. Review the performance requirements for Module C.	Verify the performance of Module C.	Not Started	Module C performs optimally under load.		
			2. Determine performance metrics to be measured.					
			3. Develop performance test cases for Module C.					
	TC004	Module D - Usability Test	1. Review the usability requirements for Module D.	Verify the usability of Module D.	Not Started	Module D is user-friendly and intuitive.		
			2. Identify usability aspects to be tested.					

EXPLANATION:

An inventory control system is critical for managing stock and supplies. Proper testing ensures the system functions accurately and efficiently.

RESULT:

Upon completion of the experiment, you will have a well-structured test plan and a set of test cases ready for testing the Inventory Control System.

EX.NO: 5	**Execute the test cases against a client server or desktop application and identify the defects.**

AIM:

The aim of this experiment is to execute the test cases against a client-server or desktop application and identify defects.

ALGORITHM:

1. **Test Environment Setup:** Set up the testing environment, including the client-server or desktop application, required hardware, and test data.
2. **Test Case Execution:** Execute the test cases designed in Experiment 2 against the application.
3. **Defect Identification:** During test execution, record any discrepancies or issues encountered.
4. **Defect Reporting:** Log the identified defects with detailed information, including steps to reproduce, severity, and priority.
5. **Defect Tracking:** Track the progress of defect resolution and verify fixes as they are implemented.
6. **Retesting:** After defect fixes, retest the affected areas to ensure the issues are resolved.
7. **Regression Testing:** Conduct regression testing to ensure new changes do not introduce new defects.

Test Case Table:

Process	No.	Test Case	Steps	Description	Status	Expected Result	Actual Result	Comment
Test Case Execution	TC001	User Login	1. Launch the application.	Verify user login process.	Not Started	User can successfully log in.		
			2. Enter valid login credentials.					
			3. Click on the "Login" button.					
	TC002	Data Validation	1. Access a data input form.	Verify data validation on the form.	Not Started	Invalid data shows appropriate error messages.		
			2. Enter invalid data in the form fields.					
			3. Submit the form.					
	TC003	File Upload	1. Access the file upload feature.	Verify file upload functionality.	Not Started	File is uploaded successfully.		

	TC004	Network Connectivity	1. Disconnect the network.	Verify the application's response.	Not Started	Application gracefully handles disconnection.		
			2. Attempt to perform an action requiring network access.					
	TC005	Concurrent Users	1. Simulate concurrent user sessions.	Verify application performance.	Not Started	Application performs well under load.		
			2. Perform actions simultaneously.					
	TC006	Compatibility	1. Test the application on different platforms.	Verify cross-platform compatibility.	Not Started	Application works on all specified platforms.		
			2. Execute tests on various browsers.					
	TC007	Client-Server Communication	1. Monitor network traffic between client and server.	Verify communication integrity.	Not Started	Data is correctly transmitted and received.		

EXPLANATION:

Testing a client-server or desktop application ensures its functionality across different platforms and environments.

RESULT:

Upon completion of the experiment, you will have a list of identified defects and their status after resolution for the client-server or desktop application.

Ex. No: 6	Test the performance of the E-commerce Application
Date:	

AIM:

To test the performance of the e-commerce application.

PROCEDURE:

Step 1: Open chrome: Use google chrome, a web browser.

Step 2: Visit your website: Go to the website you want to test.

Step 3: Check the websites performance:

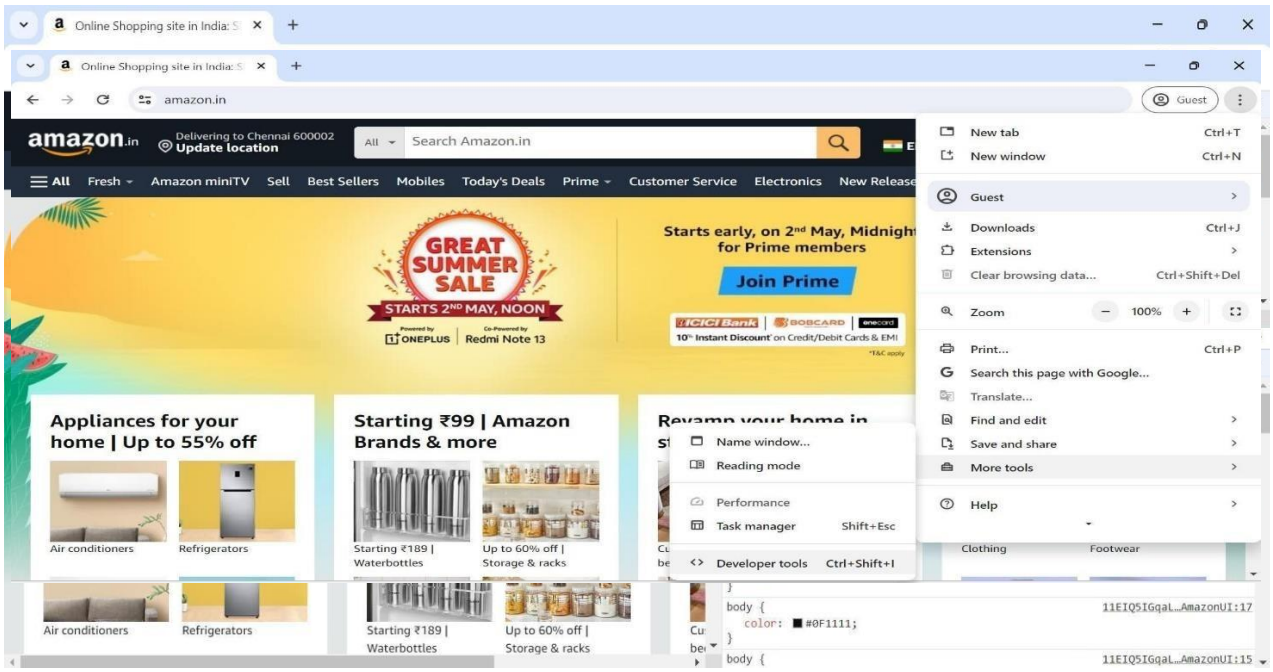
- Right click on the web page and choose "Inspect" or "Developer Tools".
- Go to the "Light house" tab it the top.

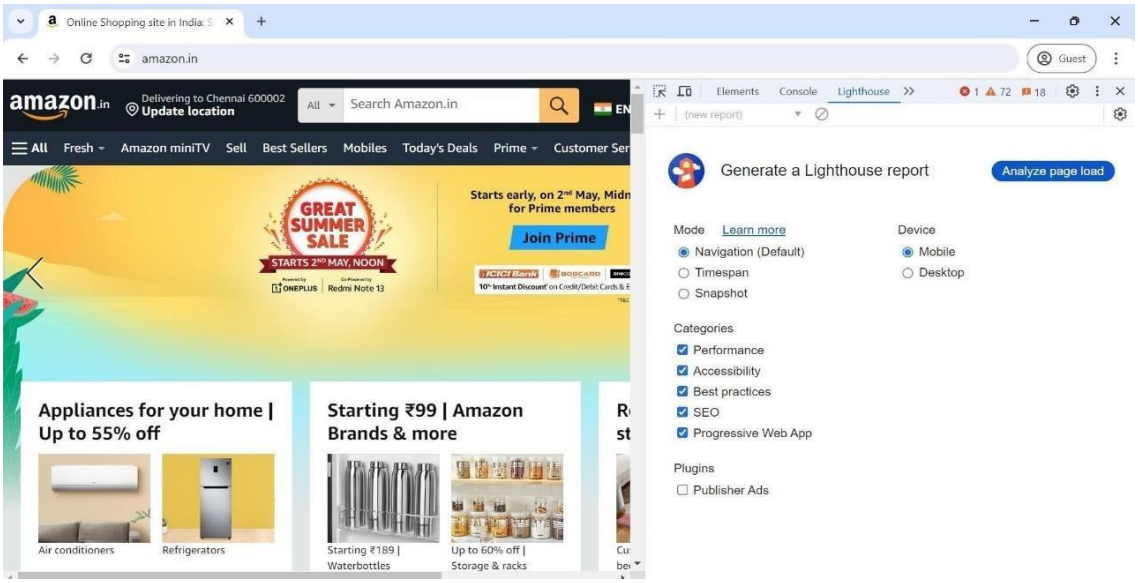
Step 4: Start the test: click on "Generate report".

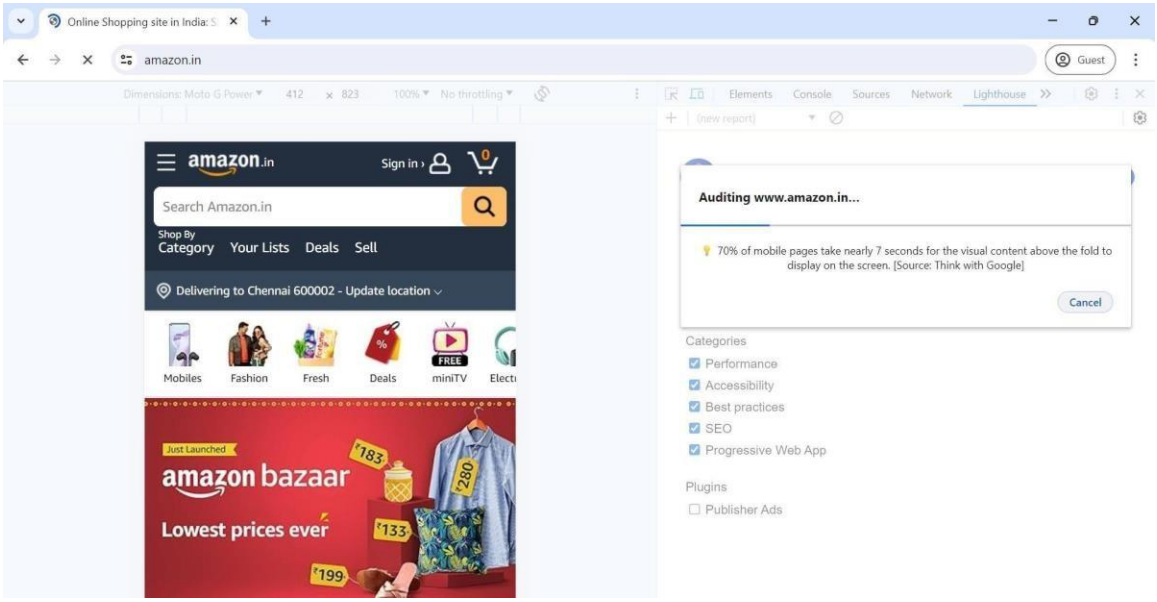
Step 5: See the result.

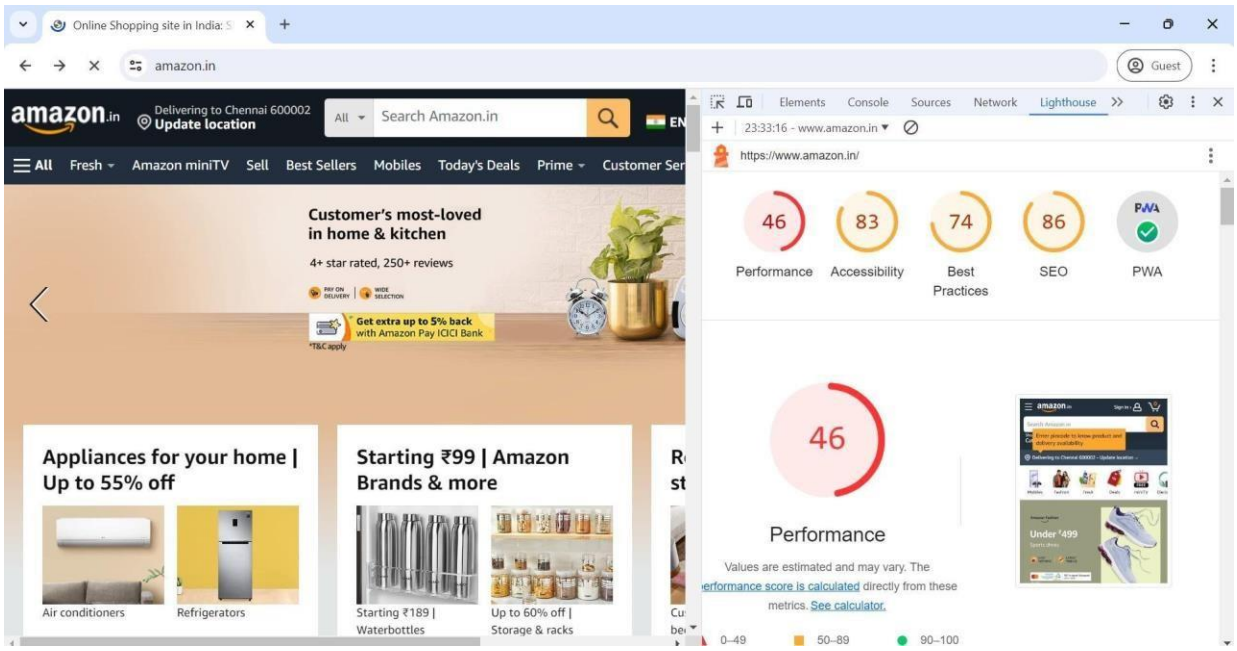
- Light house will give your website a score for performance.
- It will also provide tips to make your website faster.

Performance Test:




Online Shopping site in India: S
amazon.in
Guest
amazon.in
Delivering to Chennai 600002
Update location
All
Search Amazon.in
All
Fresh
Amazon miniTV
Sell
Best Sellers
Mobiles
Today's Deals
Prime
GREAT SUMMER SALE
STARTS 2ND MAY, NOON
Starts early, on 2nd May, Midn
for Prime members
Join Prime
Appliances for your home | Up to 55% off
Air conditioners
Refrigerators
Starting ₹99 | Amazon Brands & more
Starting ₹189 | Waterbottles
Up to 60% off | Storage & racks
Elements
Console
Lighthouse
Generate a Lighthouse report
Analyze page load
Mode
Learn more
Navigation (Default)
Timespan
Snapshot
Device
Mobile
Desktop
Categories
Performance
Accessibility
Best practices
SEO
Progressive Web App
Plugins
Publisher Ads


Online Shopping site in India: S
amazon.in
Guest
Elements
Console
Sources
Network
Lighthouse
amazon.in
Sign in
Search Amazon.in
Shop By Category
Your Lists
Deals
Sell
Delivering to Chennai 600002 - Update location
Mobiles
Fashion
Fresh
Deals
miniTV
Just Launched
amazon bazaar
Lowest prices ever
Auditing www.amazon.in...
70% of mobile pages take nearly 7 seconds for the visual content above the fold to display on the screen. [Source: Think with Google]
Cancel
Categories
Performance
Accessibility
Best practices
SEO
Progressive Web App
Plugins
Publisher Ads



RESULT:

Thus the test plan for the test performance of e-commerce applications was implemented successfully.

Ex. No: 7	THE TESTING OF E-COMMERCE APPLICATIONS USING SELENIUM
Date:	

AIM:

To automate using e-commerce application using selenium.

PROCEDURE:

Step 1: open the chrome web store then search and install the selenium IDE add on

Step 2: After installation open, selenium and give the website url

Step 3: Create a new project and give a name and rename the task name

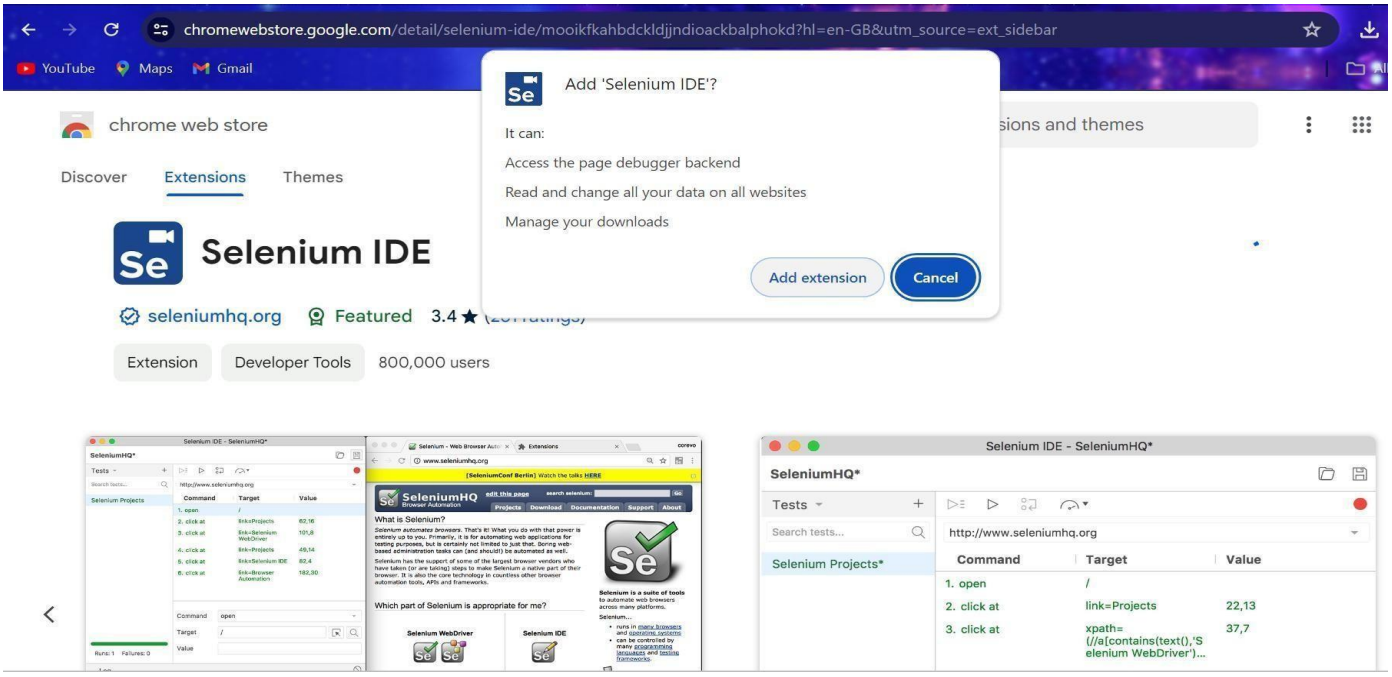
Step 4: Run url

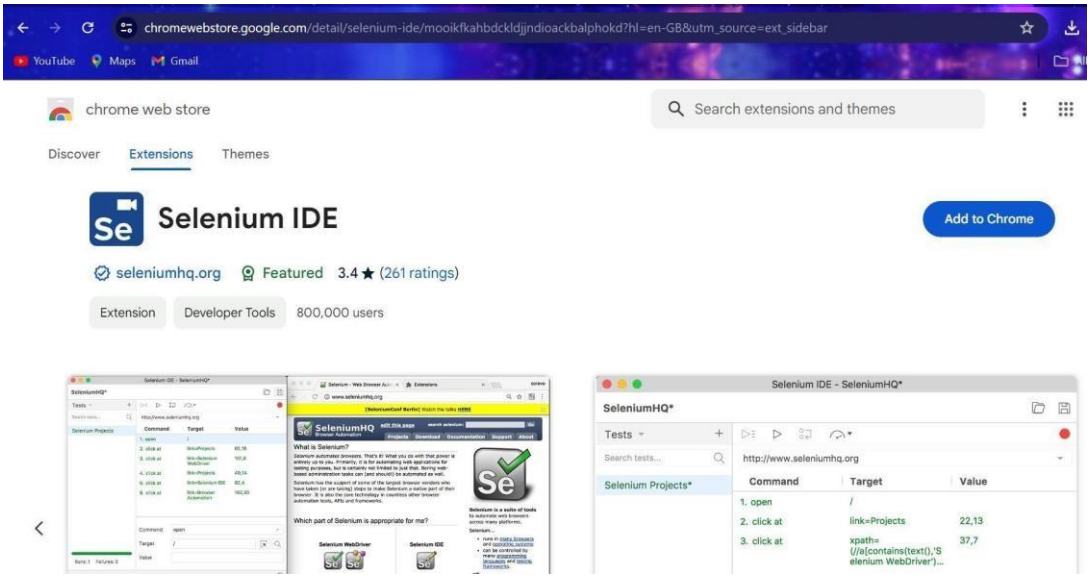
Step 5: Stop recorder

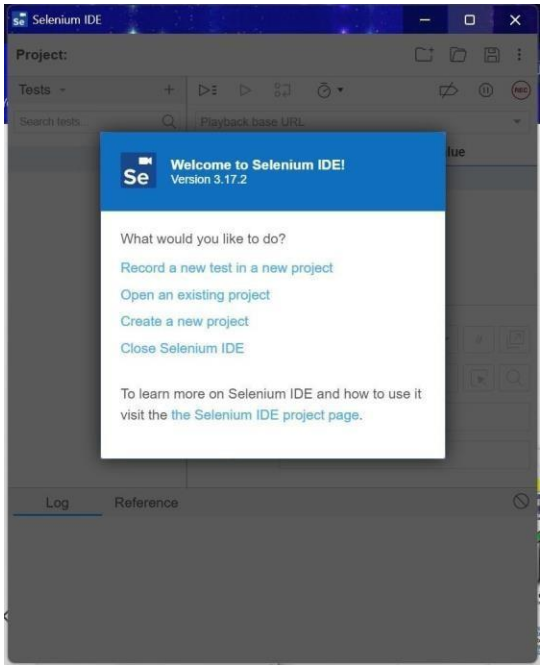
Step 6: Check if the output generated is without error

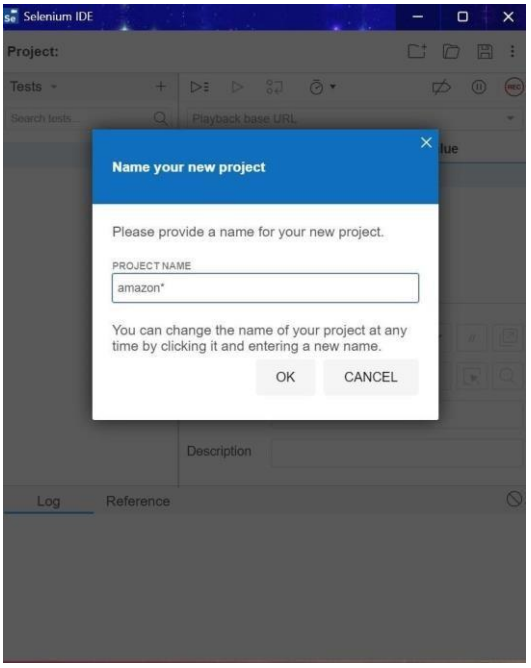
Step 7: Stop.

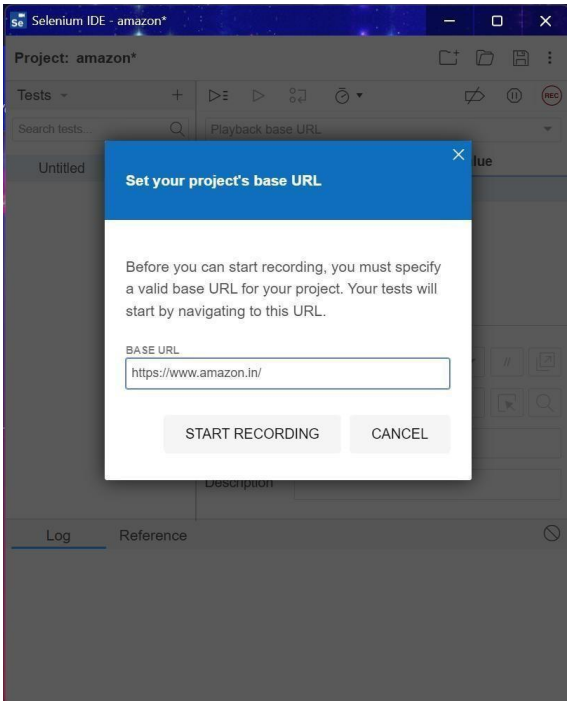
SELENIUM IDE ADD ON INSTALLATION & TESTING:



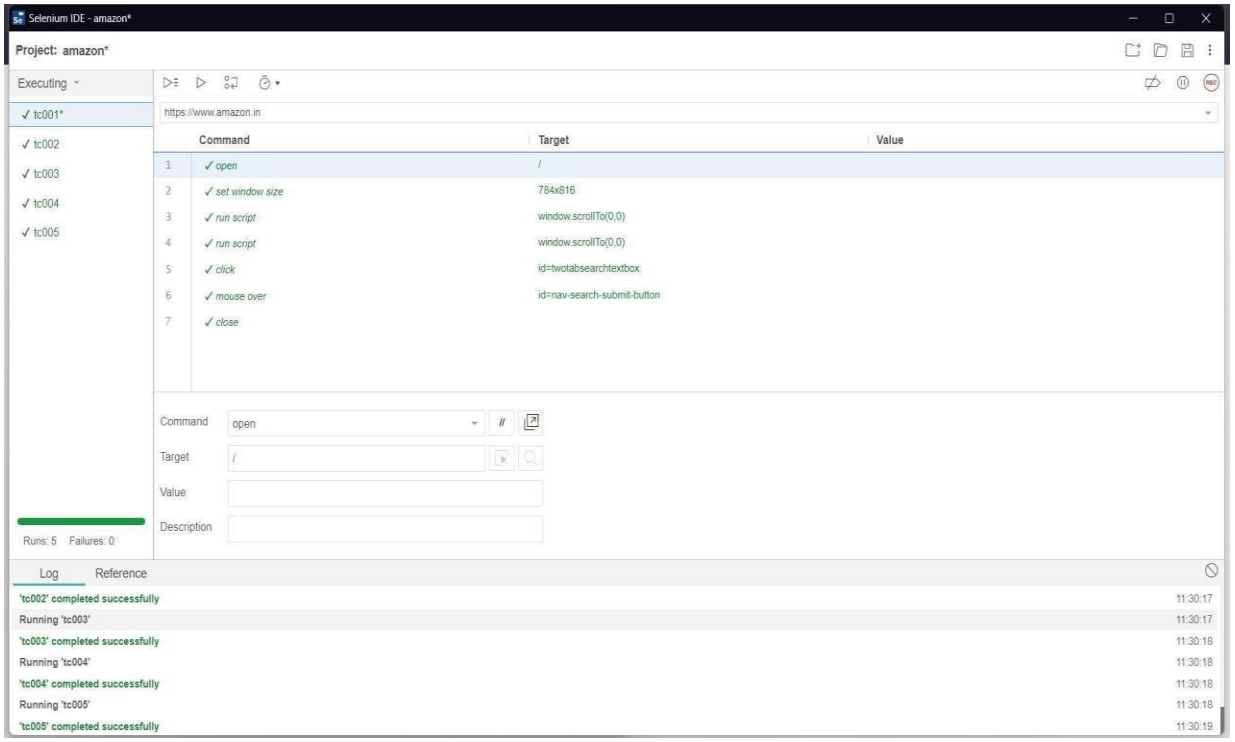

chromewebstore.google.com/detail/selenium-ide/mooikfkahbdckldjjndioackbalphokd?hl=en-GB&utm_source=ext_sidebar
YouTube
Maps
Gmail
chrome web store
Search extensions and themes
Discover
Extensions
Themes
Selenium IDE
Add to Chrome
seleniumhq.org
Featured
3.4 ★ (261 ratings)
Extension
Developer Tools
800,000 users
Selenium IDE - SeleniumHQ*
SeleniumHQ*
Tests
Search tests...
Selenium Projects*
http://www.seleniumhq.org
Command
Target
Value
1. open
/
2. click at
link=Projects
22,13
3. click at
xpath=
(//a[contains(text(),'Selenium WebDriver')...
37,7


Selenium IDE
Project:
Tests
Search tests...
Playback base URL
Welcome to Selenium IDE!
Version 3.17.2
What would you like to do?
Record a new test in a new project
Open an existing project
Create a new project
Close Selenium IDE
To learn more on Selenium IDE and how to use it visit the the Selenium IDE project page.
Log
Reference


Selenium IDE
Project:
Tests
Search tests...
Playback base URL
Name your new project
Please provide a name for your new project.
PROJECT NAME
amazon*
You can change the name of your project at any time by clicking it and entering a new name.
OK
CANCEL
Description
Log
Reference


Selenium IDE - amazon*
Project: amazon*
Tests
Search tests...
Playback base URL
Untitled
Set your project's base URL
Before you can start recording, you must specify a valid base URL for your project. Your tests will start by navigating to this URL.
BASE URL
https://www.amazon.in/
START RECORDING
CANCEL
Description
Log
Reference





RESULT:

Thus the e-commerce application is tested using selenium.

Ex. No: 8	INTEGRATE TEST NG
Date:	

AIM :

To integrate Test NG with about test automation.

PROCEDURE:

Step 1: Install Test NG in eclipse.

go to “help” >”eclipse Market place” and install Test NG.

Step 2: Create a Test NG test class.

Right -click project ,choose “new”>”class”,name it, write a Test NG test method.

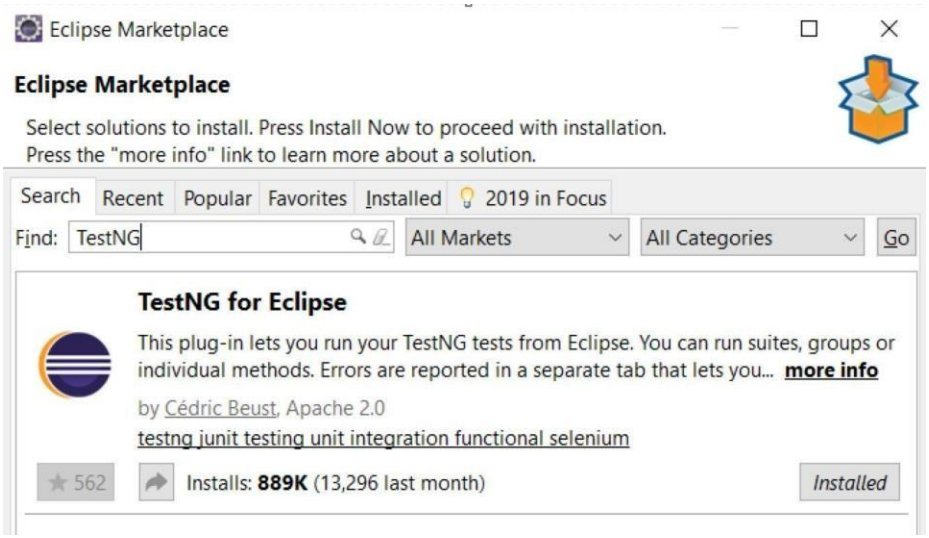
Step 3: Create “testing.xml”.

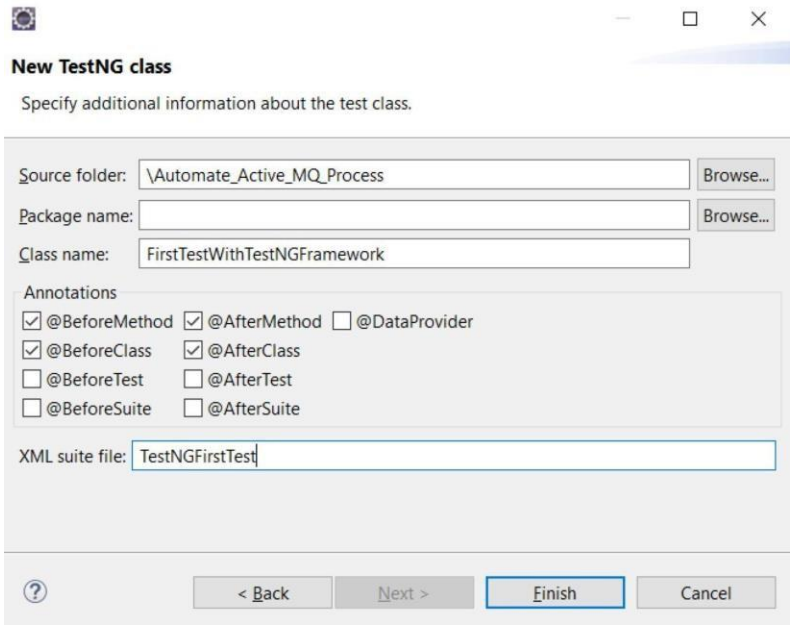
Right click project,choose “new”>”file” name it “testing.xml”,and define suite and test structure.

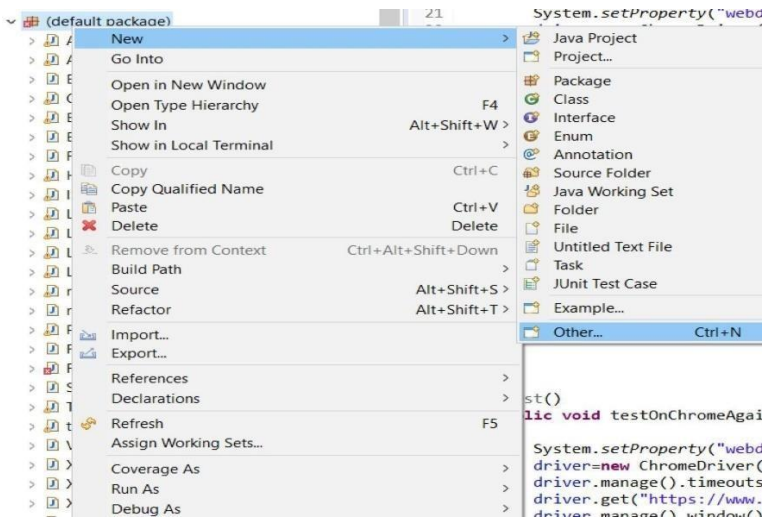
Step 4: Run Test NG suite.check eclipse console for test output.

Installation Steps & First Test using Eclipse and Test NG:






New TestNG class
Specify additional information about the test class.
Source folder: \Automate_Active_MQ_Process
Browse...
Package name:
Browse...
Class name: FirstTestWithTestNGFramework
Annotations
@BeforeMethod
@AfterMethod
@DataProvider
@BeforeClass
@AfterClass
@BeforeTest
@AfterTest
@BeforeSuite
@AfterSuite
XML suite file: TestNGFirstTest
< Back
Next >
Finish
Cancel


(default package)
New
Go Into
Open in New Window
Open Type Hierarchy F4
Show In Alt+Shift+W
Show in Local Terminal
Copy Ctrl+C
Copy Qualified Name
Paste Ctrl+V
Delete Delete
Remove from Context Ctrl+Alt+Shift+Down
Build Path
Source Alt+Shift+S
Refactor Alt+Shift+T
Import...
Export...
References
Declarations
Refresh F5
Assign Working Sets...
Coverage As
Run As
Debug As
Java Project
Project...
Package
Class
Interface
Enum
Annotation
Source Folder
Java Working Set
Folder
File
Untitled Text File
Task
JUnit Test Case
Example...
Other... Ctrl+N
System.setProperty("webd
st()
lic void testOnChromeAgai
System.setProperty("webd
driver=new ChromeDriver(
driver.manage().timeouts
driver.get("https://www.

Program:

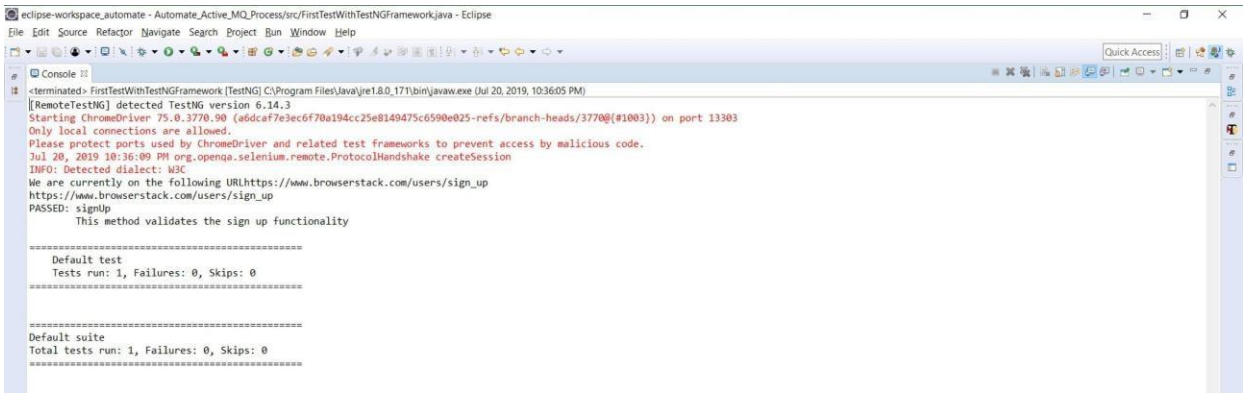
```
import org.testng.annotations.Test;
import org.testng.annotations.BeforeMethod;
import org.testng.annotations.AfterMethod;
import org.testng.annotations.BeforeClass;
import java.util.concurrent.TimeUnit;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;import
org.openqa.selenium.chrome.ChromeDriver;
import org.testng.annotations.AfterClass;

public class FirstTestWithTestNGFramework
{
WebDriver driver;
@BeforeClass
public void testSetup()
{
System.setProperty("webdriver.chrome.driver",  ".\\Driver\\chromedriver.exe");

 driver=new ChromeDriver();
driver.manage().timeouts().implicitlyWait(10, TimeUnit.SECONDS);
driver.manage().window().maximize();
}
```

```
@BeforeMethod
public void openBrowser()
{
driver.get("https://www.browserstack.com/");
driver.findElement(By.id("signupModalButton")).click();System.out.println("We are currently
on the following URL" +driver.getCurrentUrl());
}
@Test(description="This method validates the sign up functionality")
public void signUp()
{
driver.findElement(By.id("user_full_name")).sendKeys("user_name");
driver.findElement(By.id("user_email_login")).sendKeys("email_id");
driver.findElement(By.id("user_password")).sendKeys("password");
driver.findElement(By.xpath("//input[@name='terms_and_conditions']")).click();
driver.findElement(By.id("user_submit")).click();
}
@AfterMethod
public void postSignUp()
{
System.out.println(driver.getCurrentUrl());
}
@AfterClass
public void afterClass()
{
driver.quit();
}
}
```

OUTPUT :



Result:

Thus the program for integrate Test NG with about test automation was successfully verified and executed.